Android Application Development Lab

Beginner level Experiments
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Q1 -> Create an application that takes the name from a text box and shows hello message along with the name entered in text box, when the user clicks the OK button.

Ans1 ->

Xml -

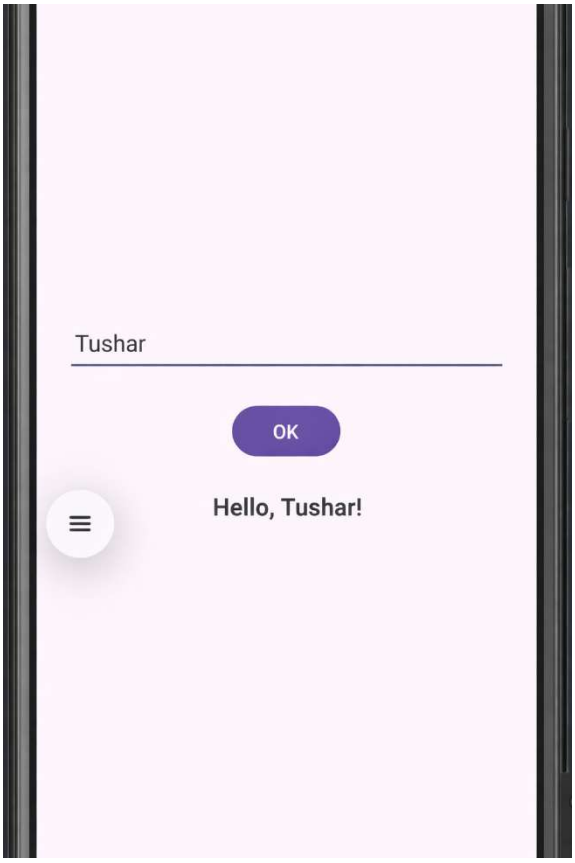
```xml
<?xml version="1.0" encoding="utf-8"?>
<LinearLayout xmlns:android="http://schemas.android.com/apk/res/android"
   android:layout_width="match_parent"
   android:layout_height="match_parent"
   android:orientation="vertical"
   android:padding="24dp"
   android:gravity="center">

   <EditText
       android:id="@+id/etName"
       android:layout_width="match_parent"
       android:layout_height="wrap_content"
       android:hint="Enter your name"
       android:inputType="textPersonName"/>

   <Button
       android:id="@+id/btnOK"
       android:layout_width="wrap_content"
       android:layout_height="wrap_content"
       android:layout_marginTop="16dp"
       android:text="OK"/>

   <TextView
       android:id="@+id/tvHello"
       android:layout_width="wrap_content"
       android:layout_height="wrap_content"
       android:layout_marginTop="24dp"
       android:textSize="20sp"
       android:textStyle="bold"
       android:text=""/>
```

```
</LinearLayout>
```

Java -

```java
package com.example.a1_user_input;

import androidx.appcompat.app.AppCompatActivity;
import android.os.Bundle;
import android.view.View;
import android.widget.Button;
import android.widget.EditText;
import android.widget.TextView;

public class MainActivity extends AppCompatActivity {

   EditText etName;
   Button btnOK;
   TextView tvHello;

   @Override
   protected void onCreate(Bundle savedInstanceState) {
       super.onCreate(savedInstanceState);
       setContentView(R.layout.activity_main);

       etName  = findViewById(R.id.etName);
       btnOK   = findViewById(R.id.btnOK);
       tvHello = findViewById(R.id.tvHello);

       btnOK.setOnClickListener(new View.OnClickListener() {
           @Override
           public void onClick(View v) {
               String name = etName.getText().toString().trim();
               if (!name.isEmpty()) {
                   tvHello.setText("Hello, " + name + "!");
               } else {
                   tvHello.setText("Please enter a name.");
               }
           }
       });
   }
}
```

Output -



Q2 -> Create a screen that has input boxes for User Name, Password, Address, Gender (radio buttons for male and female), Age (numeric), Date of Birth (Date Picket), State (Spinner) and a Submit button. On clicking the submit button, print all the data below the Submit Button. Use 1. Linear Layout, 2. Relative Lavout and 3. Grid Layout or Table Layout.

Ans2 ->

Java -

```
package com.example.formapp;

import androidx.appcompat.app.AppCompatActivity;
import android.os.Bundle;
import android.view.View;
import android.widget.*;

public class MainActivity extends AppCompatActivity {

   EditText etName, etPassword, etAddress, etAge;
   RadioGroup rgGender;
   DatePicker dpDOB;
```

```
    Spinner spState;
    Button btnSubmit;
    TextView tvResult;

    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        setContentView(R.layout.activity_main); // swap XML here

        etName     = findViewById(R.id.etName);
        etPassword = findViewById(R.id.etPassword);
        etAddress  = findViewById(R.id.etAddress);
        etAge      = findViewById(R.id.etAge);
        rgGender   = findViewById(R.id.rgGender);
        dpDOB      = findViewById(R.id.dpDOB);
        spState    = findViewById(R.id.spState);
        btnSubmit  = findViewById(R.id.btnSubmit);
        tvResult   = findViewById(R.id.tvResult);

        // Populate spinner
        String[] states = {"Delhi", "Haryana", "UP", "Maharashtra",
"Karnataka"};
        ArrayAdapter<String> adapter = new ArrayAdapter<>(this,
                android.R.layout.simple_spinner_item, states);

adapter.setDropDownViewResource(android.R.layout.simple_spinner_dropdown_item
);
        spState.setAdapter(adapter);

        btnSubmit.setOnClickListener(new View.OnClickListener() {
            @Override
            public void onClick(View v) {
                String name     = etName.getText().toString();
                String password = etPassword.getText().toString();
                String address  = etAddress.getText().toString();
                String age      = etAge.getText().toString();
                String state    = spState.getSelectedItem().toString();

                int selectedGenderId = rgGender.getCheckedRadioButtonId();
                RadioButton rbGender = findViewById(selectedGenderId);
                String gender = (rbGender != null) ?
rbGender.getText().toString() : "Not selected";

                String dob = dpDOB.getDayOfMonth() + "/" +
                        (dpDOB.getMonth() + 1) + "/" +
                        dpDOB.getYear();

                tvResult.setText(
                        "Name: "     + name     + "\n" +
```

```
                                "Password: " + password + "\n" +
                                "Address: "  + address  + "\n" +
                                "Gender: "   + gender   + "\n" +
                                "Age: "      + age      + "\n" +
                                "DOB: "      + dob      + "\n" +
                                "State: "    + state
                );
            }
        });
    }
}
```

Xml (Linear Layout) -

```
<?xml version="1.0" encoding="utf-8"?>
<ScrollView xmlns:android="http://schemas.android.com/apk/res/android"
   android:layout_width="match_parent"
   android:layout_height="match_parent">

   <LinearLayout
       android:layout_width="match_parent"
       android:layout_height="wrap_content"
       android:orientation="vertical"
       android:padding="16dp">

       <TextView android:text="User Name" android:layout_width="wrap_content"
android:layout_height="wrap_content"/>
       <EditText android:id="@+id/etName" android:layout_width="match_parent"
android:layout_height="wrap_content" android:hint="Enter name"/>

       <TextView android:text="Password" android:layout_width="wrap_content"
android:layout_height="wrap_content" android:layout_marginTop="8dp"/>
       <EditText android:id="@+id/etPassword"
android:layout_width="match_parent" android:layout_height="wrap_content"
android:inputType="textPassword" android:hint="Enter password"/>

       <TextView android:text="Address" android:layout_width="wrap_content"
android:layout_height="wrap_content" android:layout_marginTop="8dp"/>
       <EditText android:id="@+id/etAddress"
android:layout_width="match_parent" android:layout_height="wrap_content"
android:hint="Enter address"/>

       <TextView android:text="Gender" android:layout_width="wrap_content"
android:layout_height="wrap_content" android:layout_marginTop="8dp"/>
       <RadioGroup android:id="@+id/rgGender"
android:orientation="horizontal" android:layout_width="wrap_content"
android:layout_height="wrap_content">
```

```
            <RadioButton android:id="@+id/rbMale" android:text="Male"
android:layout_width="wrap_content" android:layout_height="wrap_content"/>
            <RadioButton android:id="@+id/rbFemale" android:text="Female"
android:layout_width="wrap_content" android:layout_height="wrap_content"/>
        </RadioGroup>

        <TextView android:text="Age" android:layout_width="wrap_content"
android:layout_height="wrap_content" android:layout_marginTop="8dp"/>
        <EditText android:id="@+id/etAge" android:layout_width="match_parent"
android:layout_height="wrap_content" android:inputType="number"
android:hint="Enter age"/>

        <TextView android:text="Date of Birth"
android:layout_width="wrap_content" android:layout_height="wrap_content"
android:layout_marginTop="8dp"/>
        <DatePicker android:id="@+id/dpDOB"
android:layout_width="wrap_content" android:layout_height="wrap_content"
android:calendarViewShown="false"/>

        <TextView android:text="State" android:layout_width="wrap_content"
android:layout_height="wrap_content" android:layout_marginTop="8dp"/>
        <Spinner android:id="@+id/spState" android:layout_width="match_parent"
android:layout_height="wrap_content"/>

        <Button android:id="@+id/btnSubmit"
android:layout_width="match_parent" android:layout_height="wrap_content"
android:layout_marginTop="16dp" android:text="Submit"/>

        <TextView android:id="@+id/tvResult"
android:layout_width="wrap_content" android:layout_height="wrap_content"
android:layout_marginTop="16dp" android:textSize="15sp"/>

    </LinearLayout>
</ScrollView>
```

Output -

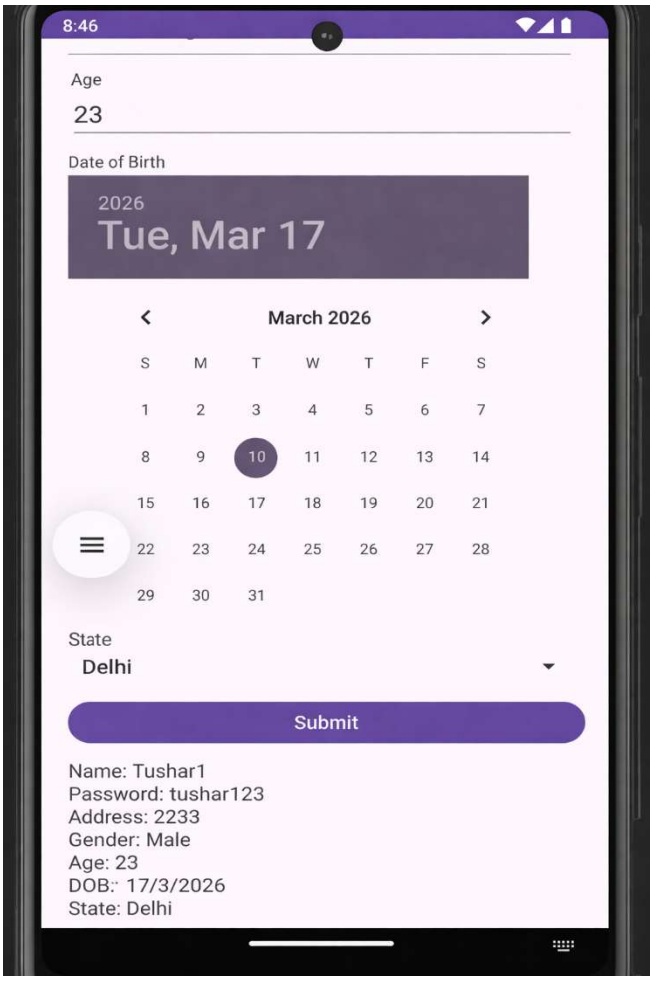
Xml (Relative Layout) -

```
<?xml version="1.0" encoding="utf-8"?>
<ScrollView xmlns:android="http://schemas.android.com/apk/res/android"
   android:layout_width="match_parent"
   android:layout_height="match_parent">

   <RelativeLayout
       android:layout_width="match_parent"
       android:layout_height="wrap_content"
       android:padding="16dp">

       <TextView android:id="@+id/lblName" android:text="User Name"
           android:layout_width="wrap_content"
android:layout_height="wrap_content"/>
       <EditText android:id="@+id/etName" android:hint="Enter name"
           android:layout_width="match_parent"
android:layout_height="wrap_content"
           android:layout_below="@id/lblName"/>

       <TextView android:id="@+id/lblPassword" android:text="Password"
           android:layout_width="wrap_content"
android:layout_height="wrap_content"
           android:layout_below="@id/etName" android:layout_marginTop="8dp"/>
```

```
        <EditText android:id="@+id/etPassword" android:hint="Enter password"
            android:inputType="textPassword"
            android:layout_width="match_parent"
android:layout_height="wrap_content"
            android:layout_below="@id/lblPassword"/>

        <TextView android:id="@+id/lblAddress" android:text="Address"
            android:layout_width="wrap_content"
android:layout_height="wrap_content"
            android:layout_below="@id/etPassword"
android:layout_marginTop="8dp"/>
        <EditText android:id="@+id/etAddress" android:hint="Enter address"
            android:layout_width="match_parent"
android:layout_height="wrap_content"
            android:layout_below="@id/lblAddress"/>

        <TextView android:id="@+id/lblGender" android:text="Gender"
            android:layout_width="wrap_content"
android:layout_height="wrap_content"
            android:layout_below="@id/etAddress"
android:layout_marginTop="8dp"/>
        <RadioGroup android:id="@+id/rgGender"
android:orientation="horizontal"
            android:layout_width="wrap_content"
android:layout_height="wrap_content"
            android:layout_below="@id/lblGender">
            <RadioButton android:id="@+id/rbMale" android:text="Male"
                android:layout_width="wrap_content"
android:layout_height="wrap_content"/>
            <RadioButton android:id="@+id/rbFemale" android:text="Female"
                android:layout_width="wrap_content"
android:layout_height="wrap_content"/>
        </RadioGroup>

        <TextView android:id="@+id/lblAge" android:text="Age"
            android:layout_width="wrap_content"
android:layout_height="wrap_content"
            android:layout_below="@id/rgGender"
android:layout_marginTop="8dp"/>
        <EditText android:id="@+id/etAge" android:hint="Enter age"
            android:inputType="number"
            android:layout_width="match_parent"
android:layout_height="wrap_content"
            android:layout_below="@id/lblAge"/>

        <TextView android:id="@+id/lblDOB" android:text="Date of Birth"
            android:layout_width="wrap_content"
android:layout_height="wrap_content"
            android:layout_below="@id/etAge" android:layout_marginTop="8dp"/>
```

```
        <DatePicker android:id="@+id/dpDOB"
            android:layout_width="wrap_content"
android:layout_height="wrap_content"
            android:calendarViewShown="false"
            android:layout_below="@id/lblDOB"/>

        <TextView android:id="@+id/lblState" android:text="State"
            android:layout_width="wrap_content"
android:layout_height="wrap_content"
            android:layout_below="@id/dpDOB" android:layout_marginTop="8dp"/>
        <Spinner android:id="@+id/spState"
            android:layout_width="match_parent"
android:layout_height="wrap_content"
            android:layout_below="@id/lblState"/>

        <Button android:id="@+id/btnSubmit" android:text="Submit"
            android:layout_width="match_parent"
android:layout_height="wrap_content"
            android:layout_below="@id/spState"
android:layout_marginTop="16dp"/>

        <TextView android:id="@+id/tvResult"
            android:layout_width="wrap_content"
android:layout_height="wrap_content"
            android:layout_below="@id/btnSubmit"
android:layout_marginTop="16dp"
            android:textSize="15sp"/>

    </RelativeLayout>
</ScrollView>
```

Output -



Xml (Table Layout) -

```xml
<?xml version="1.0" encoding="utf-8"?>
<ScrollView xmlns:android="http://schemas.android.com/apk/res/android"
   android:layout_width="match_parent"
   android:layout_height="match_parent">

   <TableLayout
       android:layout_width="match_parent"
       android:layout_height="wrap_content"
       android:padding="16dp"
       android:stretchColumns="1">

       <TableRow>
           <TextView android:text="User Name" android:padding="8dp"/>
           <EditText android:id="@+id/etName" android:hint="Enter name"/>
       </TableRow>

       <TableRow>
           <TextView android:text="Password" android:padding="8dp"/>
```

```
            <EditText android:id="@+id/etPassword" android:hint="Enter
password" android:inputType="textPassword"/>
        </TableRow>

        <TableRow>
            <TextView android:text="Address" android:padding="8dp"/>
            <EditText android:id="@+id/etAddress" android:hint="Enter
address"/>
        </TableRow>

        <TableRow>
            <TextView android:text="Gender" android:padding="8dp"/>
            <RadioGroup android:id="@+id/rgGender"
android:orientation="horizontal">
                <RadioButton android:id="@+id/rbMale" android:text="Male"/>
                <RadioButton android:id="@+id/rbFemale"
android:text="Female"/>
            </RadioGroup>
        </TableRow>

        <TableRow>
            <TextView android:text="Age" android:padding="8dp"/>
            <EditText android:id="@+id/etAge" android:hint="Enter age"
android:inputType="number"/>
        </TableRow>

        <TableRow>
            <TextView android:text="DOB" android:padding="8dp"/>
            <DatePicker android:id="@+id/dpDOB"
android:calendarViewShown="false"/>
        </TableRow>

        <TableRow>
            <TextView android:text="State" android:padding="8dp"/>
            <Spinner android:id="@+id/spState"/>
        </TableRow>

        <TableRow>
            <Button android:id="@+id/btnSubmit" android:text="Submit"
                android:layout_span="2" android:layout_width="match_parent"/>
        </TableRow>

        <TableRow>
            <TextView android:id="@+id/tvResult" android:layout_span="2"
                android:padding="8dp" android:textSize="15sp"/>
        </TableRow>

    </TableLayout>
</ScrollView>
```

Output -

Q3 -> Develop an application that inserts some notifications into Notification area and whenever a notification is inserted, it should show a toast with details of the notification.

Ans3 ->

Xml -

```
<?xml version="1.0" encoding="utf-8"?>
<LinearLayout xmlns:android="http://schemas.android.com/apk/res/android"
   android:layout_width="match_parent"
   android:layout_height="match_parent"
   android:orientation="vertical"
   android:gravity="center"
   android:padding="24dp">

   <EditText
       android:id="@+id/etTitle"
       android:layout_width="match_parent"
       android:layout_height="wrap_content"
       android:hint="Notification Title"/>
```

```
    <EditText
        android:id="@+id/etMessage"
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:hint="Notification Message"
        android:layout_marginTop="12dp"/>

    <Button
        android:id="@+id/btnSend"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:layout_marginTop="20dp"
        android:text="Send Notification"/>

</LinearLayout>
```

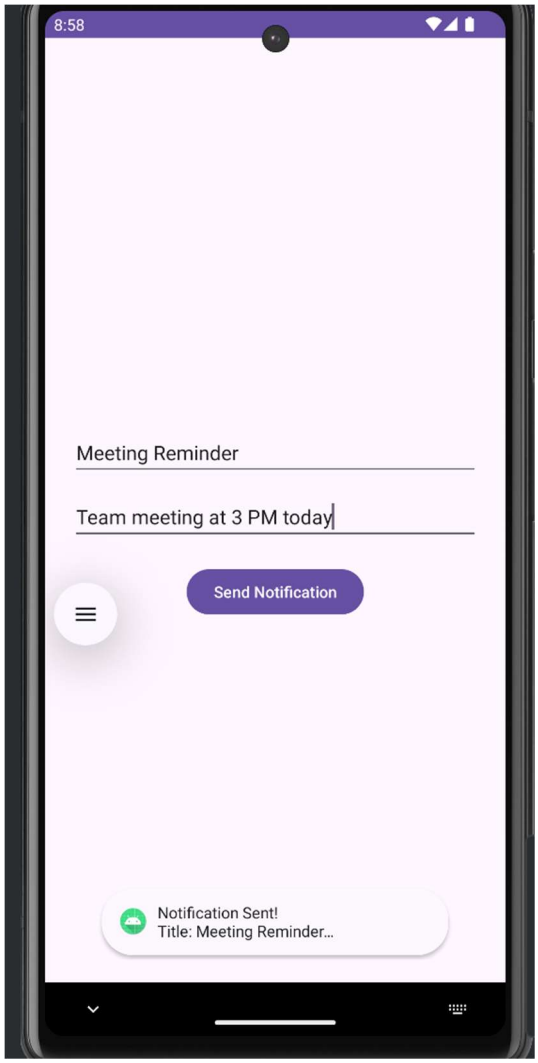
Java -

```
package com.example.q3notification;

import androidx.appcompat.app.AppCompatActivity;
import androidx.core.app.NotificationCompat;
import androidx.core.app.NotificationManagerCompat;
import android.app.NotificationChannel;
import android.app.NotificationManager;
import android.os.Build;
import android.os.Bundle;
import android.view.View;
import android.widget.Button;
import android.widget.EditText;
import android.widget.Toast;

public class MainActivity extends AppCompatActivity {

    EditText etTitle, etMessage;
    Button btnSend;

    private static final String CHANNEL_ID = "my_channel";
    private static int notificationId = 0;

    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        setContentView(R.layout.activity_main);

        etTitle   = findViewById(R.id.etTitle);
        etMessage = findViewById(R.id.etMessage);
        btnSend   = findViewById(R.id.btnSend);
```

```
        createNotificationChannel();

        btnSend.setOnClickListener(new View.OnClickListener() {
            @Override
            public void onClick(View v) {
                String title   = etTitle.getText().toString().trim();
                String message = etMessage.getText().toString().trim();

                if (title.isEmpty() || message.isEmpty()) {
                    Toast.makeText(MainActivity.this,
                            "Please enter title and message",
Toast.LENGTH_SHORT).show();
                    return;
                }

                // Build and show notification
                NotificationCompat.Builder builder = new
NotificationCompat.Builder(
                        MainActivity.this, CHANNEL_ID)
                        .setSmallIcon(android.R.drawable.ic_dialog_info)
                        .setContentTitle(title)
                        .setContentText(message)
                        .setPriority(NotificationCompat.PRIORITY_HIGH)
                        .setAutoCancel(true);

                NotificationManagerCompat notificationManager =
                        NotificationManagerCompat.from(MainActivity.this);
                notificationManager.notify(notificationId++, builder.build());

                // Show Toast with notification details
                Toast.makeText(MainActivity.this,
                        "Notification Sent!\nTitle: " + title + "\nMessage: "
+ message,
                        Toast.LENGTH_LONG).show();
            }
        });
    }

    private void createNotificationChannel() {
        if (Build.VERSION.SDK_INT >= Build.VERSION_CODES.O) {
            CharSequence name        = "My Channel";
            String description       = "Channel for app notifications";
            int importance           = NotificationManager.IMPORTANCE_HIGH;
            NotificationChannel channel = new NotificationChannel(CHANNEL_ID,
name, importance);
            channel.setDescription(description);

            NotificationManager notificationManager =
                    getSystemService(NotificationManager.class);
```

```
            notificationManager.createNotificationChannel(channel);
        }
    }
}
```

Output -



Q4 -> Develop an application that uses a menu with 3 options for dialing a number, opening a website and to send an SMS. On selecting an option, the appropriate action should be invoked using intents.

Ans4 ->

Xml -

```
<?xml version="1.0" encoding="utf-8"?>
<LinearLayout xmlns:android="http://schemas.android.com/apk/res/android"
```

```
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    android:orientation="vertical">

    <androidx.appcompat.widget.Toolbar
        android:id="@+id/toolbar"
        android:layout_width="match_parent"
        android:layout_height="?attr/actionBarSize"
        android:background="?attr/colorPrimary"
        android:title="Q4 Menu App"/>

    <TextView
        android:layout_width="match_parent"
        android:layout_height="match_parent"
        android:gravity="center"
        android:text="Press the 3-dot menu (top right)"
        android:textSize="18sp"/>

</LinearLayout>
```

Java -

```
package com.example.q4menu;

import androidx.appcompat.app.AppCompatActivity;
import android.content.Intent;
import android.net.Uri;
import android.os.Bundle;
import android.view.Menu;
import android.view.MenuItem;
import android.widget.Toast;
import androidx.appcompat.widget.Toolbar;

public class MainActivity extends AppCompatActivity {

    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        setContentView(R.layout.activity_main);
        Toolbar toolbar = findViewById(R.id.toolbar);
        setSupportActionBar(toolbar);
    }

    // Inflate the menu
    @Override
    public boolean onCreateOptionsMenu(Menu menu) {
        getMenuInflater().inflate(R.menu.menu_main, menu);
        return true;
    }
```

```java
    // Handle menu item clicks
    @Override
    public boolean onOptionsItemSelected(MenuItem item) {
        int id = item.getItemId();

        if (id == R.id.menu_dial) {
            // Dial a number
            Intent dialIntent = new Intent(Intent.ACTION_DIAL);
            dialIntent.setData(Uri.parse("tel:9876543210"));
            startActivity(dialIntent);
            return true;

        } else if (id == R.id.menu_website) {
            // Open a website
            Intent webIntent = new Intent(Intent.ACTION_VIEW);
            webIntent.setData(Uri.parse("https://www.google.com"));
            startActivity(webIntent);
            return true;

        } else if (id == R.id.menu_sms) {
            // Send an SMS
            Intent smsIntent = new Intent(Intent.ACTION_SENDTO);
            smsIntent.setData(Uri.parse("smsto:9876543210"));
            smsIntent.putExtra("sms_body", "Hello! This is a test SMS.");
            startActivity(smsIntent);
            return true;
        }

        return super.onOptionsItemSelected(item);
    }
}
```

Output -

Q5 -> Create an application that uses a text file to store user names and passwords (tab separated fields and one record per line) When the user submits a login name and password through a screen, the details should be verified with the text file data and if they match, show a dialog saying that login is successful. Otherwise, show the dialog with Login Failed message.

Ans5 ->

Xml -

```
<?xml version="1.0" encoding="utf-8"?>
<LinearLayout xmlns:android="http://schemas.android.com/apk/res/android"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    android:orientation="vertical"
    android:gravity="center"
    android:padding="24dp">

    <TextView
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
```

```
        android:text="Login"
        android:textSize="28sp"
        android:textStyle="bold"
        android:layout_marginBottom="24dp"/>

    <EditText
        android:id="@+id/etUsername"
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:hint="Enter Username"
        android:inputType="textPersonName"/>

    <EditText
        android:id="@+id/etPassword"
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:hint="Enter Password"
        android:inputType="textPassword"
        android:layout_marginTop="12dp"/>

    <Button
        android:id="@+id/btnLogin"
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:layout_marginTop="20dp"
        android:text="Login"/>

    <Button
        android:id="@+id/btnRegister"
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:layout_marginTop="8dp"
        android:text="Register (Save User)"/>

</LinearLayout>
```

Java -

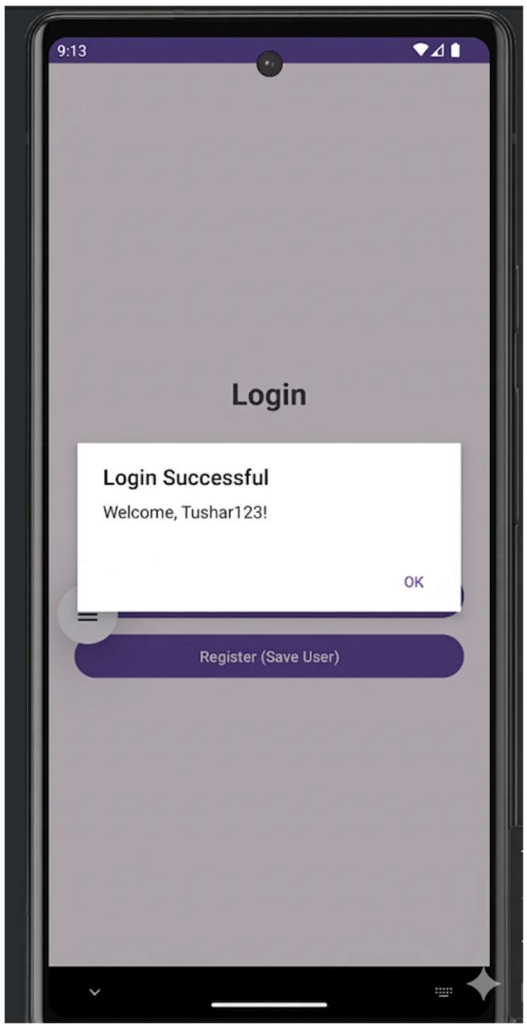
```
package com.example.q5login;

import androidx.appcompat.app.AlertDialog;
import androidx.appcompat.app.AppCompatActivity;
import android.os.Bundle;
import android.view.View;
import android.widget.Button;
import android.widget.EditText;
import android.widget.Toast;
import java.io.*;
```

```java
public class MainActivity extends AppCompatActivity {

   EditText etUsername, etPassword;
   Button btnLogin, btnRegister;
   private static final String FILE_NAME = "users.txt";

   @Override
   protected void onCreate(Bundle savedInstanceState) {
       super.onCreate(savedInstanceState);
       setContentView(R.layout.activity_main);

       etUsername  = findViewById(R.id.etUsername);
       etPassword  = findViewById(R.id.etPassword);
       btnLogin    = findViewById(R.id.btnLogin);
       btnRegister = findViewById(R.id.btnRegister);

       // Save new user to text file
       btnRegister.setOnClickListener(new View.OnClickListener() {
           @Override
           public void onClick(View v) {
               String username = etUsername.getText().toString().trim();
               String password = etPassword.getText().toString().trim();

               if (username.isEmpty() || password.isEmpty()) {
                   Toast.makeText(MainActivity.this,
                           "Please enter username and password",
Toast.LENGTH_SHORT).show();
                   return;
               }

               try {
                   FileOutputStream fos = openFileOutput(FILE_NAME,
MODE_APPEND);
                   String record = username + "\t" + password + "\n";
                   fos.write(record.getBytes());
                   fos.close();
                   Toast.makeText(MainActivity.this,
                           "User registered successfully!",
Toast.LENGTH_SHORT).show();
               } catch (IOException e) {
                   e.printStackTrace();
               }
           }
       });

       // Verify login from text file
       btnLogin.setOnClickListener(new View.OnClickListener() {
           @Override
           public void onClick(View v) {
```

```
                String username = etUsername.getText().toString().trim();
                String password = etPassword.getText().toString().trim();

                if (username.isEmpty() || password.isEmpty()) {
                    Toast.makeText(MainActivity.this,
                            "Please enter username and password",
Toast.LENGTH_SHORT).show();
                    return;
                }

                boolean loginSuccess = false;

                try {
                    FileInputStream fis = openFileInput(FILE_NAME);
                    BufferedReader reader = new BufferedReader(new
InputStreamReader(fis));
                    String line;

                    while ((line = reader.readLine()) != null) {
                        String[] parts = line.split("\t");
                        if (parts.length == 2) {
                            String savedUser = parts[0].trim();
                            String savedPass = parts[1].trim();
                            if (savedUser.equals(username) &&
savedPass.equals(password)) {
                                loginSuccess = true;
                                break;
                            }
                        }
                    }
                    reader.close();
                    fis.close();

                } catch (FileNotFoundException e) {
                    showDialog("Login Failed", "No users registered yet!");
                    return;
                } catch (IOException e) {
                    e.printStackTrace();
                }

                if (loginSuccess) {
                    showDialog("Login Successful", "Welcome, " + username +
"!");
                } else {
                    showDialog("Login Failed", "Invalid username or
password.");
                }
            }
        });
```

```
    }

    private void showDialog(String title, String message) {
        new AlertDialog.Builder(this)
                .setTitle(title)
                .setMessage(message)
                .setPositiveButton("OK", null)
                .show();
    }
}
```

Output -



Q6 -> Create an activity to navigate to different activities through Intents-Implicit and Explicit.

Ans6 ->

Xml -

```xml
<?xml version="1.0" encoding="utf-8"?>
<LinearLayout xmlns:android="http://schemas.android.com/apk/res/android"
   android:layout_width="match_parent"
   android:layout_height="match_parent"
   android:orientation="vertical"
   android:gravity="center"
   android:padding="24dp">

   <TextView
       android:layout_width="wrap_content"
       android:layout_height="wrap_content"
       android:text="Intent Navigation Demo"
       android:textSize="22sp"
       android:textStyle="bold"
       android:layout_marginBottom="32dp"/>

   <Button
       android:id="@+id/btnExplicit"
       android:layout_width="match_parent"
       android:layout_height="wrap_content"
       android:text="Explicit Intent (Go to Second Activity)"/>

   <Button
       android:id="@+id/btnImplicitWeb"
       android:layout_width="match_parent"
       android:layout_height="wrap_content"
       android:layout_marginTop="12dp"
       android:text="Implicit Intent (Open Website)"/>

   <Button
       android:id="@+id/btnImplicitCall"
       android:layout_width="match_parent"
       android:layout_height="wrap_content"
       android:layout_marginTop="12dp"
       android:text="Implicit Intent (Dial Number)"/>

   <Button
       android:id="@+id/btnImplicitMap"
       android:layout_width="match_parent"
       android:layout_height="wrap_content"
       android:layout_marginTop="12dp"
       android:text="Implicit Intent (Open Map)"/>

</LinearLayout>
```

Xml (acitivity_second.xml) -

```xml
<?xml version="1.0" encoding="utf-8"?>
<LinearLayout xmlns:android="http://schemas.android.com/apk/res/android"
   android:layout_width="match_parent"
   android:layout_height="match_parent"
   android:orientation="vertical"
   android:gravity="center"
   android:padding="24dp">

   <TextView
       android:id="@+id/tvMessage"
       android:layout_width="wrap_content"
       android:layout_height="wrap_content"
       android:text="Welcome to Second Activity!"
       android:textSize="22sp"
       android:textStyle="bold"/>

   <Button
       android:id="@+id/btnBack"
       android:layout_width="wrap_content"
       android:layout_height="wrap_content"
       android:layout_marginTop="24dp"
       android:text="Go Back"/>

</LinearLayout>
```

Java (MainActivity.java) -

```java
package com.example.q6intents;

import androidx.appcompat.app.AppCompatActivity;
import android.content.Intent;
import android.net.Uri;
import android.os.Bundle;
import android.view.View;
import android.widget.Button;

public class MainActivity extends AppCompatActivity {

   Button btnExplicit, btnImplicitWeb, btnImplicitCall, btnImplicitMap;

   @Override
   protected void onCreate(Bundle savedInstanceState) {
       super.onCreate(savedInstanceState);
       setContentView(R.layout.activity_main);

       btnExplicit    = findViewById(R.id.btnExplicit);
       btnImplicitWeb = findViewById(R.id.btnImplicitWeb);
       btnImplicitCall= findViewById(R.id.btnImplicitCall);
       btnImplicitMap = findViewById(R.id.btnImplicitMap);
```

```
        // EXPLICIT INTENT — go to SecondActivity
        btnExplicit.setOnClickListener(new View.OnClickListener() {
            @Override
            public void onClick(View v) {
                Intent intent = new Intent(MainActivity.this,
SecondActivity.class);
                intent.putExtra("message", "Hello from MainActivity!");
                startActivity(intent);
            }
        });

        // IMPLICIT INTENT — open website
        btnImplicitWeb.setOnClickListener(new View.OnClickListener() {
            @Override
            public void onClick(View v) {
                Intent intent = new Intent(Intent.ACTION_VIEW);
                intent.setData(Uri.parse("https://www.google.com"));
                startActivity(intent);
            }
        });

        // IMPLICIT INTENT — dial number
        btnImplicitCall.setOnClickListener(new View.OnClickListener() {
            @Override
            public void onClick(View v) {
                Intent intent = new Intent(Intent.ACTION_DIAL);
                intent.setData(Uri.parse("tel:9876543210"));
                startActivity(intent);
            }
        });

        // IMPLICIT INTENT — open map
        btnImplicitMap.setOnClickListener(new View.OnClickListener() {
            @Override
            public void onClick(View v) {
                Intent intent = new Intent(Intent.ACTION_VIEW);
                intent.setData(Uri.parse("geo:28.6139,77.2090?q=New+Delhi"));
                startActivity(intent);
            }
        });
    }
}
```

Java (SecondActivity.java) -

```
package com.example.q6intents;

import androidx.appcompat.app.AppCompatActivity;
import android.os.Bundle;
```

```java
import android.view.View;
import android.widget.Button;
import android.widget.TextView;

public class SecondActivity extends AppCompatActivity {

    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        setContentView(R.layout.activity_second);

        TextView tvMessage = findViewById(R.id.tvMessage);
        Button btnBack     = findViewById(R.id.btnBack);

        // Receive data from MainActivity
        String message = getIntent().getStringExtra("message");
        if (message != null) {
            tvMessage.setText(message);
        }

        btnBack.setOnClickListener(new View.OnClickListener() {
            @Override
            public void onClick(View v) {
                finish(); // goes back to MainActivity
            }
        });
    }
}
```

Output -

Q7 -> Create an activity to create option Menu.
Ans7 ->

Xml (activity_main.xml) -

```
<?xml version="1.0" encoding="utf-8"?>
<LinearLayout xmlns:android="http://schemas.android.com/apk/res/android"
   android:layout_width="match_parent"
   android:layout_height="match_parent"
   android:orientation="vertical"
   android:gravity="center"
   android:padding="24dp">

   <TextView
       android:id="@+id/tvResult"
       android:layout_width="wrap_content"
       android:layout_height="wrap_content"
       android:text="Select an option from the menu"
       android:textSize="20sp"
       android:gravity="center"
       android:textStyle="bold"/>

</LinearLayout>
```

Xml (menu_options.xml) -

```xml
<?xml version="1.0" encoding="utf-8"?>
<menu xmlns:android="http://schemas.android.com/apk/res/android">

   <item
       android:id="@+id/menu_home"
       android:title="Home"
       android:icon="@android:drawable/ic_menu_today"
       android:showAsAction="ifRoom"/>

   <item
       android:id="@+id/menu_search"
       android:title="Search"
       android:icon="@android:drawable/ic_menu_search"
       android:showAsAction="ifRoom"/>

   <item
       android:id="@+id/menu_settings"
       android:title="Settings"
       android:icon="@android:drawable/ic_menu_preferences"
       android:showAsAction="never"/>

   <item
       android:id="@+id/menu_share"
       android:title="Share"
       android:icon="@android:drawable/ic_menu_share"
       android:showAsAction="never"/>

   <item
       android:id="@+id/menu_about"
       android:title="About"
       android:icon="@android:drawable/ic_menu_info_details"
       android:showAsAction="never"/>

   <item
       android:id="@+id/menu_exit"
       android:title="Exit"
       android:icon="@android:drawable/ic_menu_close_clear_cancel"
       android:showAsAction="never"/>

</menu>
```

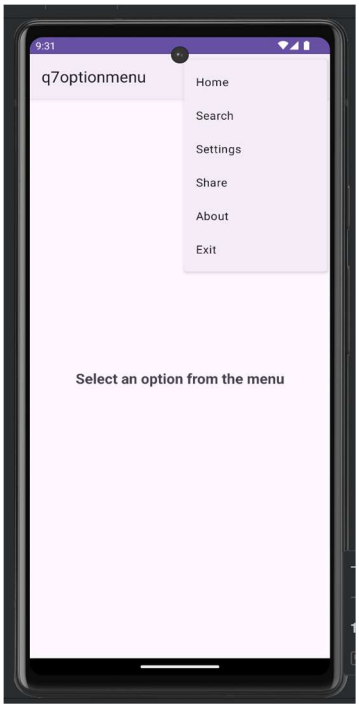
Java -

```java
package com.example.q7optionmenu;

import androidx.appcompat.app.AlertDialog;
import androidx.appcompat.app.AppCompatActivity;
```

```java
import android.os.Bundle;
import android.view.Menu;
import android.view.MenuItem;
import android.widget.TextView;
import android.widget.Toast;

public class MainActivity extends AppCompatActivity {

    TextView tvResult;

    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        setContentView(R.layout.activity_main);

        tvResult = findViewById(R.id.tvResult);
    }

    // Inflate the menu
    @Override
    public boolean onCreateOptionsMenu(Menu menu) {
        getMenuInflater().inflate(R.menu.menu_options, menu);
        return true;
    }

    // Handle menu item clicks
    @Override
    public boolean onOptionsItemSelected(MenuItem item) {
        int id = item.getItemId();

        if (id == R.id.menu_home) {
            tvResult.setText("You selected: Home");
            Toast.makeText(this, "Home selected", Toast.LENGTH_SHORT).show();
            return true;

        } else if (id == R.id.menu_search) {
            tvResult.setText("You selected: Search");
            Toast.makeText(this, "Search selected",
Toast.LENGTH_SHORT).show();
            return true;

        } else if (id == R.id.menu_settings) {
            tvResult.setText("You selected: Settings");
            Toast.makeText(this, "Settings selected",
Toast.LENGTH_SHORT).show();
            return true;

        } else if (id == R.id.menu_share) {
            tvResult.setText("You selected: Share");
```

```
            Toast.makeText(this, "Share selected", Toast.LENGTH_SHORT).show();
            return true;

        } else if (id == R.id.menu_about) {
            // Show About dialog
            new AlertDialog.Builder(this)
                    .setTitle("About")
                    .setMessage("This is Question 7 - Option Menu
App\nDeveloped for Android Lab")
                    .setPositiveButton("OK", null)
                    .show();
            return true;

        } else if (id == R.id.menu_exit) {
            // Show exit confirmation dialog
            new AlertDialog.Builder(this)
                    .setTitle("Exit")
                    .setMessage("Are you sure you want to exit?")
                    .setPositiveButton("Yes", (dialog, which) -> finish())
                    .setNegativeButton("No", null)
                    .show();
            return true;
        }

        return super.onOptionsItemSelected(item);
    }
}
```

Output -



Q8 -> Create an activity using Button Controls- CheckBox, RadioButtons, Radio Groups
Ans8 ->

Xml -

```
<?xml version="1.0" encoding="utf-8"?>
<ScrollView xmlns:android="http://schemas.android.com/apk/res/android"
   android:layout_width="match_parent"
   android:layout_height="match_parent">

   <LinearLayout
       android:layout_width="match_parent"
       android:layout_height="wrap_content"
       android:orientation="vertical"
       android:padding="24dp">

       <!-- CHECKBOXES -->
       <TextView
           android:layout_width="wrap_content"
           android:layout_height="wrap_content"
           android:text="Select Hobbies (CheckBox):"
           android:textSize="18sp"
           android:textStyle="bold"/>

       <CheckBox
           android:id="@+id/cbReading"
           android:layout_width="wrap_content"
           android:layout_height="wrap_content"
           android:text="Reading"/>
```

```
        <CheckBox
            android:id="@+id/cbCoding"
            android:layout_width="wrap_content"
            android:layout_height="wrap_content"
            android:text="Coding"/>

        <CheckBox
            android:id="@+id/cbGaming"
            android:layout_width="wrap_content"
            android:layout_height="wrap_content"
            android:text="Gaming"/>

        <CheckBox
            android:id="@+id/cbTravelling"
            android:layout_width="wrap_content"
            android:layout_height="wrap_content"
            android:text="Travelling"/>

        <!-- RADIO BUTTONS -->
        <TextView
            android:layout_width="wrap_content"
            android:layout_height="wrap_content"
            android:text="Select Gender (RadioButton):"
            android:textSize="18sp"
            android:textStyle="bold"
            android:layout_marginTop="20dp"/>

        <RadioGroup
            android:id="@+id/rgGender"
            android:layout_width="wrap_content"
            android:layout_height="wrap_content"
            android:orientation="horizontal">

            <RadioButton
                android:id="@+id/rbMale"
                android:layout_width="wrap_content"
                android:layout_height="wrap_content"
                android:text="Male"/>

            <RadioButton
                android:id="@+id/rbFemale"
                android:layout_width="wrap_content"
                android:layout_height="wrap_content"
                android:text="Female"/>

            <RadioButton
                android:id="@+id/rbOther"
                android:layout_width="wrap_content"
                android:layout_height="wrap_content"
```

```
            android:text="Other"/>

    </RadioGroup>

    <!-- RADIO GROUP — Age Range -->
    <TextView
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:text="Select Age Range (RadioGroup):"
        android:textSize="18sp"
        android:textStyle="bold"
        android:layout_marginTop="20dp"/>

    <RadioGroup
        android:id="@+id/rgAge"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:orientation="vertical">

        <RadioButton
            android:id="@+id/rbAge1"
            android:layout_width="wrap_content"
            android:layout_height="wrap_content"
            android:text="Below 18"/>

        <RadioButton
            android:id="@+id/rbAge2"
            android:layout_width="wrap_content"
            android:layout_height="wrap_content"
            android:text="18 - 25"/>

        <RadioButton
            android:id="@+id/rbAge3"
            android:layout_width="wrap_content"
            android:layout_height="wrap_content"
            android:text="26 - 35"/>

        <RadioButton
            android:id="@+id/rbAge4"
            android:layout_width="wrap_content"
            android:layout_height="wrap_content"
            android:text="Above 35"/>

    </RadioGroup>

    <!-- SUBMIT BUTTON -->
    <Button
        android:id="@+id/btnSubmit"
        android:layout_width="match_parent"
```

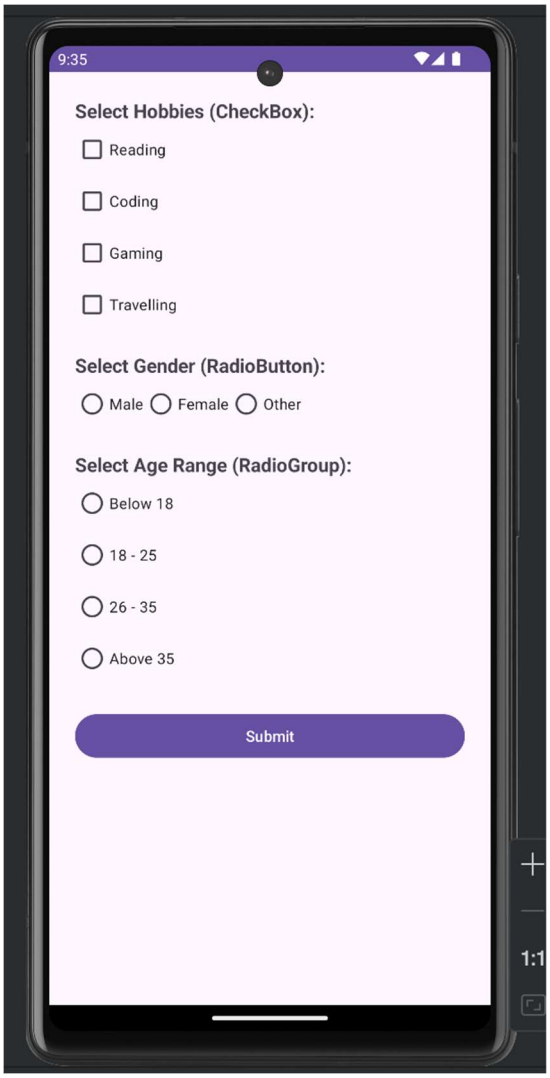
```
            android:layout_height="wrap_content"
            android:layout_marginTop="24dp"
            android:text="Submit"/>

        <!-- RESULT -->
        <TextView
            android:id="@+id/tvResult"
            android:layout_width="wrap_content"
            android:layout_height="wrap_content"
            android:layout_marginTop="16dp"
            android:textSize="16sp"
            android:textStyle="bold"/>

    </LinearLayout>
</ScrollView>
```

Java -

```
package com.example.q8controls;

import androidx.appcompat.app.AppCompatActivity;
import android.os.Bundle;
import android.view.View;
import android.widget.*;

public class MainActivity extends AppCompatActivity {

    CheckBox cbReading, cbCoding, cbGaming, cbTravelling;
    RadioGroup rgGender, rgAge;
    Button btnSubmit;
    TextView tvResult;

    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        setContentView(R.layout.activity_main);

        cbReading    = findViewById(R.id.cbReading);
        cbCoding     = findViewById(R.id.cbCoding);
        cbGaming     = findViewById(R.id.cbGaming);
        cbTravelling = findViewById(R.id.cbTravelling);
        rgGender     = findViewById(R.id.rgGender);
        rgAge        = findViewById(R.id.rgAge);
        btnSubmit    = findViewById(R.id.btnSubmit);
        tvResult     = findViewById(R.id.tvResult);

        btnSubmit.setOnClickListener(new View.OnClickListener() {
            @Override
            public void onClick(View v) {
```

```java
                // Get selected hobbies
                String hobbies = "";
                if (cbReading.isChecked())    hobbies += "Reading, ";
                if (cbCoding.isChecked()) hobbies += "Coding, ";
                if (cbGaming.isChecked()) hobbies += "Gaming, ";
                if (cbTravelling.isChecked()) hobbies += "Travelling, ";
                if (hobbies.isEmpty()) hobbies = "None";
                else hobbies = hobbies.substring(0, hobbies.length() - 2);

                // Get selected gender
                int genderId = rgGender.getCheckedRadioButtonId();
                String gender = "Not selected";
                if (genderId != -1) {
                    RadioButton rb = findViewById(genderId);
                    gender = rb.getText().toString();
                }

                // Get selected age range
                int ageId = rgAge.getCheckedRadioButtonId();
                String ageRange = "Not selected";
                if (ageId != -1) {
                    RadioButton rb = findViewById(ageId);
                    ageRange = rb.getText().toString();
                }

                // Display result
                tvResult.setText(
                        "Hobbies: "   + hobbies  + "\n" +
                                "Gender: "    + gender   + "\n" +
                                "Age Range: " + ageRange
                );
            }
        });
    }
}
```

Output -


9:35
Select Hobbies (CheckBox):
Reading
Coding
Gaming
Travelling
Select Gender (RadioButton):
Male
Female
Other
Select Age Range (RadioGroup):
Below 18
18 - 25
26 - 35
Above 35
Submit
1:1